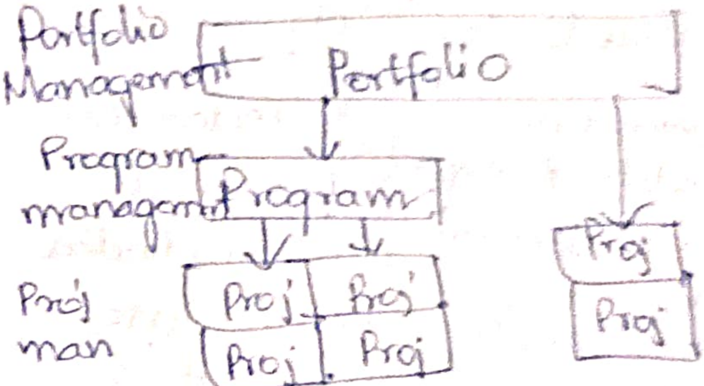
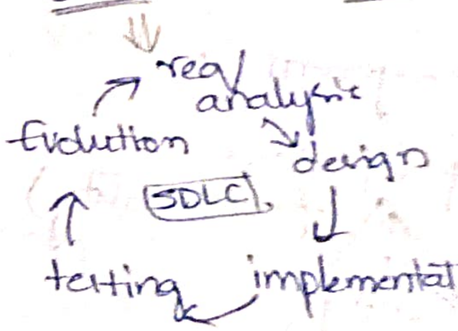
24/7/25

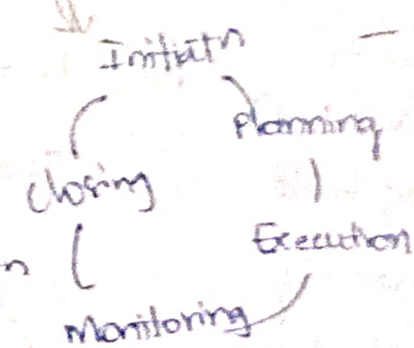
11) What s/w PM's will do?
- managing res
- risk management
- Quality control
- Communicatn c̄ in & outside of team & stakeholders
- Budget management
- Timeline management
-

Portfolio Vs Program Vs Proj



SDLC Vs Proj life cycle





Sw dev Projs – Success Vs failure

	2011	2012	2013	2014	2015
Success	29	27	31	28	29
challenged	49	56	50	55	52
failure	22	17	19	17	19

Success – on time, on budget, most of the fun has implemented

challenge – if ~~an~~ one of above 3 is affected.

failed if any of 3 things hatted (or) unused after delivering

02/08/2025

Activity

objectives, buy vs build & Process model selection

1. Proj vs Proj management obj (acti)
2. Build vs buy decisions
3. Process model Selection
4. Plan driven process models
5. Agile methods.

Project?

- a complex, non-routine, one-time effort; limited by time, " budget, " resources, & perf specifications designed to meet customer needs.
- a proj is unique, has specific time frame (not a continuing/ongoing activity like automobile productn..)

>req >src, interdependent activities & have degree of uncertainity.

Proj Stakeholders:

- who have stake/interest in proj.
- they could be users/clients, dev/implementers
- they could be
  - in Proj team
  - outside of team in same Org.
  - outside of team & Org.
- diff stakeholders may have diff obj.

Proj characteristics -

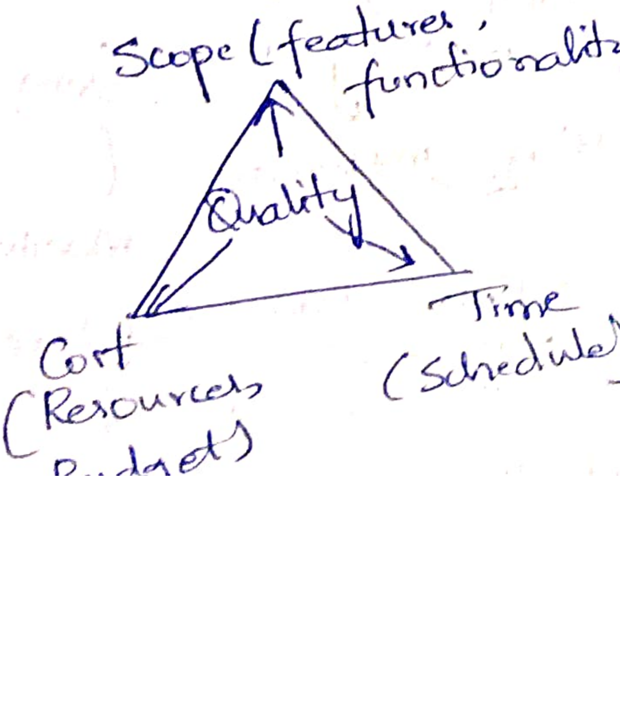
- Risks & uncertainities -
  eg: 1. are req well understood?
  2. are technologies to be used well understood?
- Type of appln being built
  eg: 1. info sys/embedded sys?
  2. criticality vs diff bw target & dev environments!

Proj obj Vs Proj manage obj

1. define desired outcomes, focusing on end prod/sys & its alignment with bu goals
2. describes what proj aims to achieve in terms of functionality, perf & value to Org.
3. Centered on product & its impact on st, dev team & bu.

1. process of delivering the proj.
2. how proj will be managed to achieve proj obj.
3'. Centered on proj executn, including timel, budgets, resrc, risks & st coordination.

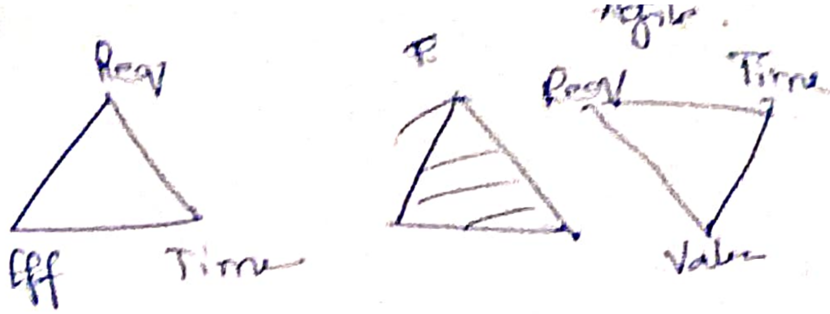
Proj Management Δ^{le}



Build vs Buy Decision:

3 Options:

1. Buy a COTS soln
2. Build In-house
3. Build c̄ a 3rd party Vendor.



1. Buy a COTS soln.

- Purchasing Pre-built sw frm Vendor that meets proj functional req (eg: CRM, ERP, industry specific systems.
  Implementatn involves licensing, Configuring to fit org needs & integratng c̄ existing System.
  Vendors provide cloud-based/ on-premise options, along c̄ ongng support & updates.
- Key Benefits:
  - Fast deployment (3-6mnts),
  - lower initial costs &
  - reduced dev risk due to proven soln.
  - Ideal for std req & org c̄ limited tech resrc.
- Key Drawbacks:
  - Limited Customizatn, potentially requiring workarounds for unique needs, maintainance fee, dependency on Vendor support

Proj goals & obj & Proj charter

7/5/25

1. Goals vs obj
2. Proj goals & Proj manag goals.
3. Mapping goals to smart obj
4. Preparing Proj charter.

aspect	goals	~~obj~~ obj
def	broad, general intentions / outcomes to be achieved.	specific, measurable steps / actions to achieve goals
Nature	Qualitative, high-level.	Quantitative, detailed.
time frame	long-term	short to medium
measurability	often hard to measure directly.	Always measurable.
Scope	strategic - align with vision / bu drivers	Tactical - tied to deliverables & milestones.
who uses	sponsors, executives, cust	Proj managers, team mem.
eg:	"Enhance cust exp"	"Reduce avg online ticket booking time by 30% in 3 mnts"

Proj goals vs Proj manag goals

Proj goals	Proj manag goals
cust exp & serve	on-time delivery, staying in budget
high level outcomes.	successful exec & delivery of Proj
align with broader bu obj	manage proj constraints like time scope

2. Mapping goals to SMART obj.

1. Specific: clearly defined.
2. Measurable: Quantifiable to track
3. Achievable: realistic given resrc.
4. Relevant: aligned c̄ proj priorities
5. Time bound: tied to deadline

// SMART is a frm to ensure obj are clear & actionable.

eg:
Cust Satisfactn can be measured by fb survey, CSAT & NPS.

// activities:

Proj charter:
- a foundational doc that officially authorizes the existence of a proj & gives PM the authority to use org resrcs to execute proj.

Key Purpose
- Formally authorise the proj
- Align all stks on proj obj, scope & purpose.
- Establish PM's authority
- serves as a reference point throughout proj lifecycle.

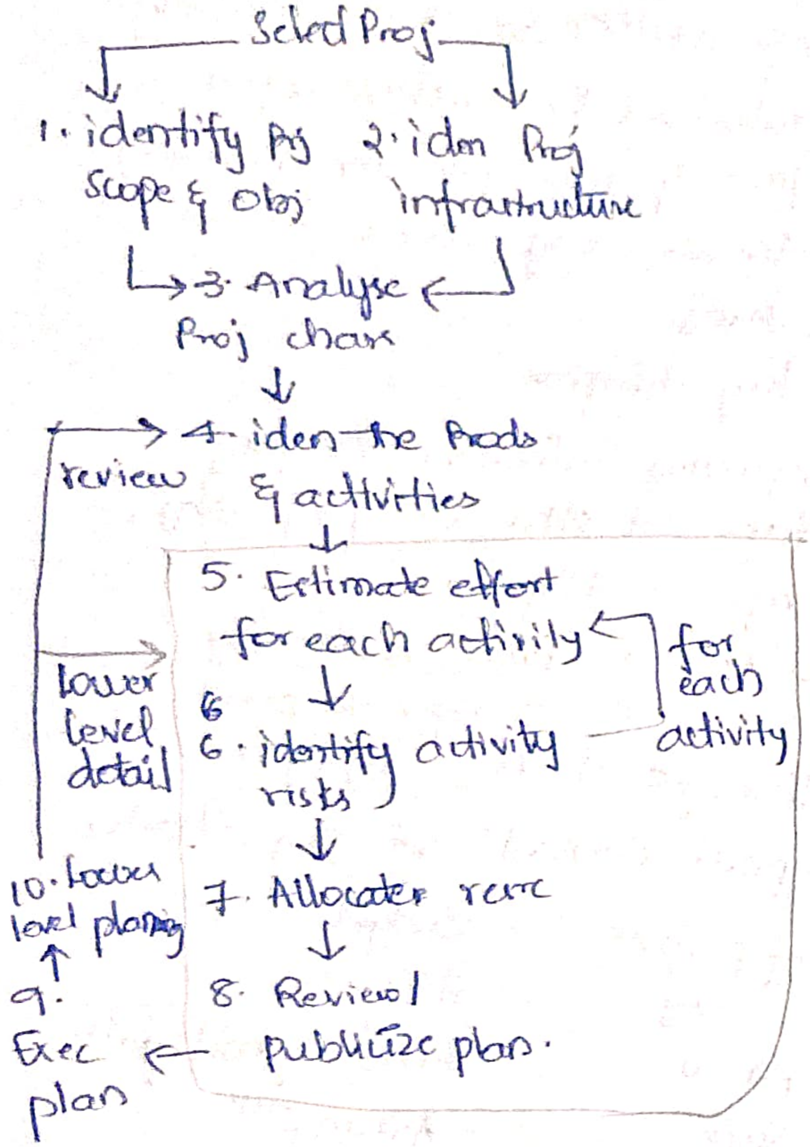
Typical contents of Proj charter.

- Proj purpose
- " obj
- PM "
- Scope
- Exectn approach
- Key deliverables
- Timeline
- Budget
- Stks
- Risks & mitigatn
- Assumptns
- Constraints
- Approval.

Proj charter Comparison

	Plan-driven	agile
Purpose	define entire scope, t/l, deliv upfront	align on prod vision, goals, team expectn
Detail level	high	Lean
change flexibility	Low	High
ownership	Sponsor + PM driven	Prod owner + Team driven c̄ PM facilitatn
planning horizon	full proj duratn	curr release (or) ↑ (1-3) mnths
use in exectn	reference for contract, fixed deliv	alignment tool for iteratns / sprints

Stepwise Proj Planning (Overview) : [opt]



16/8/2025

Prod & Process measures + Effort Estimatn

1. Prod/process & direct/Indirect Measures.

	direct measures	Indirect measures
Prod metrics	1. Lines of code (LOC / KLOC) 2. Cyclomatic Complexity 3. Defect Density	1. Cust satisfaction (surveys [NPS) 2. Maintainability Index (ex, complexity)
Proj metrics	1. Earned Value 2. Cycle Time	1. Dev Productivity 2. Risk Exposure.

Direct Vs Indirect measures.

Direct	Indirect
- quantifiable.	- Qualitative/Quantitative
- involve straight-fwd data collectn	- involve interpretatn/ inference
- focus on measurable aspects like code, time /cost	- focus on outcomes/ effects that are influenced by multiple factors.

Act: Connecting goals, obj's, metrics & functional specificatns.

Size - Oriented metrics.

- metrics are derived by normalizing quality &/or prod measures by considering size of sw produced in KLOC (1000s lines of code)

Eg metrics.

- Errors per KLOC [per person-month)
- Defects per KLOC]
- Cost per KLOC (Cost per page of doctn)

KLOC Measures.

- dependent on Prog lang.
- Penalize well designed but short Pograms.
- Can't easily accomodate.

Function - Oriented metrics.

- functionality delivered by appln as a normalized va

  - Errors per FP
  - Defects " "
  - Cost " "
  - Pages of doctn per FP
  - FP per person-month

Use of Metrics.

- Enables PM to
  - access status of ongoing Proj - tackle potential ri
  - uncover problm before they go critical
  - adjust work flow
  - Evaluate team ability

20 Person months :
One person can complete Proj in 20 months (or)
20 ppl in 1 month (or)
10 ppl in 2 months

Effort esti - approaches

- Expert judgement
- Analogous est
- Parametric "
- Bottom - up "
- Top - down
- wideband Delphi tech
- function points
- Use - case points
- COCOMO (Constructive Cost Model)
- Story Points (planning Poker)

FP tech

- each req. is mapped to several functions & logical data entities / files
- Calculatn of size is based on functions supported by sw & data files used by sw.
- functions are of 3 types
  - External Inputs
  - " Outputs
  - " Queries
- Data files : 2 types
  - Internal logic files
  - External ~~logic~~ Interface files,

Types of FP Analysis

Transactional FP
- External ip
- External op
- External Inquiries

Data FP
- Internal logical file
- Exte inte file

FP for functions & files :-

Program char	low Complexity	medium	high
E I/P	_x3	_x4	_x6
E O/P	_x4	_x5	_x7
E Queries	_x3	_x4	_x6
Int logical files	_x4	_x10	_x15
Ex Interface files	_x5	_x7	_x10

Unadjust FP (UFPs) = sum of all

St1: Identify function Types.
no. of EI
EO
EQ
ILF
EIF

St2: Determine f Compl

St3: Sum of wt = unadjusted F.P

St3: Apply Value adjustment Factor (VAF)
- there will complexities for som deried from ratings.
CF

St4: ~~FC~~

$$\boxed{\text{Adjusted FP} = \text{unadjusted FP} \times \text{Value Adjust Fact}}$$

Effort = $\dfrac{\text{Adjusted FP}}{\text{FP per person}}$

≈ ____ Person months

VAF 14 Factors

1. Data Comm^n
2. Distributed Data Processing
3. Performance
4. Heavily Used Config
5. Transaction Rate
6. On-line Data Entry
7. End User Efficiency
8. Online Update
9. Complex processing
10. Reusability
11. Installat^n Ease
12. Operat^nal "
13. multiple sites
14. Facilitate change

Q) Consider a s/w Proj c̄ below info:
No. of External i/p = 30 (I)
" o/p = 60 (O)
" Inquiries = 23 (E)
Files (F) = 8
External interfaces (N) = 02.

Given corresponding complexity wt factor
I O E F N (4, 5, 4, 10, 7)
[Simple].

also given that, out of 14 value adjustment factors that infl dev effort, 4 factors are not applicable. each of other 4 fac has value 3 and each of remaining fact has value 4.
Computed value of FP is?

Sol $\boxed{FP = UFP \times VAF}$

i) UFP = Unadjusted FP.

	no. of i/p	wt factor	term
EI	30	4	120
EO	60	5	300
EQ	23	4	92
IF	8	10	80
IN	02.	7	14

~~VAF~~

UFP. ~~VAF~~ = 606.

ii) VAF.
degree of influence is calculated.
(14 char factors are there. it shows how much these char are effecting).
14 characteristics.

$\boxed{VAF = 0.65 + [0.01 \times \Sigma GSC]}$

ΣGSC = sum of influence of 14 factors
each value varies from 0 to 5

If 1 FP cost 100
Per month budget =
17 × 100 = 17[illegible]

Given: 4 factors has no impact = 0
~~6~~4 factors " 3 impact = 4×3 = 12
rest " " 4 " = ~~4×4~~ 6×4 = ~~16~~ 24 (14-

ΣGSC = ~~16~~ 12 + 24 = 36

$VAF = 0.65 + [0.01 \times \cancel{16}\ 36] = 1.01$

$FP = UFP \times VAF = 606 \times 1.01 = 612.06.$

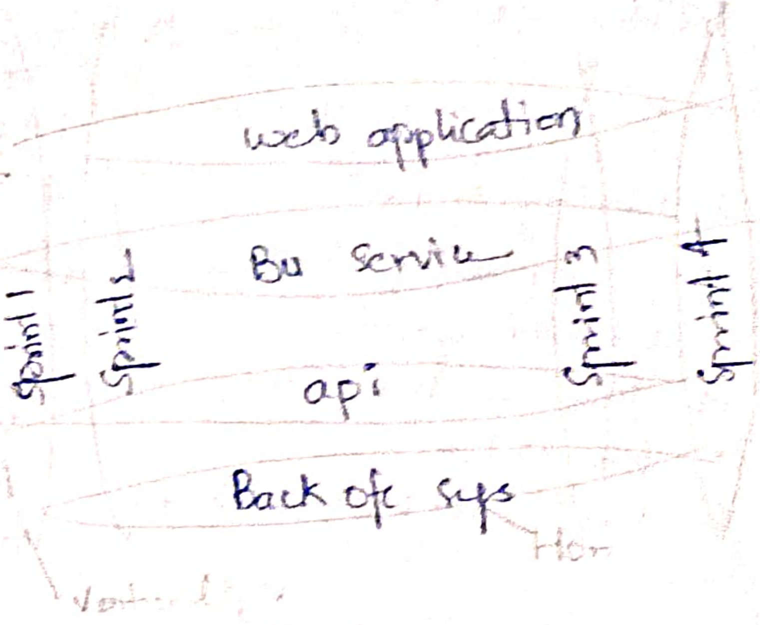
∴ If Effort = 36 person months

$\boxed{Productivity = \frac{FP}{Effort}} = \frac{612.06}{36} = 17$ FP/[illegible]

Scrum - Estimation.

traditional est : days, weeks, mnths.

Agile: fibonacci
T-shirt sizing.
Powers of 2.



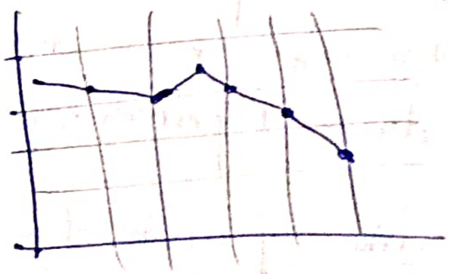
Planning Poker tech:

- all parti play cards & estimate items
- di after indiv esti discussion is raised when large diff identified
- Process is repeate till whole team reached concensus abt agreed est (which is not avg esti)

		sp1 sp2 sp3 sp4
epi 1	UI layer	ui layer
2	middle lay	m lay
3	sec lay	sec lay
4	DB lay	DB lay
	Horizontal.	Vertical.

User story decompositn (Eg):

Burndown chart - eq.



User story decomposition

horizontal vs Vertical slicing.

⇓ (mostly for tech items)
⇓ (delivering functionality across diff layers of archi)

⇓ (Business focus)
⇓ (delivering end to end funct incrementally)

Splitting US:

- Hamburger meth.

1. Identify tasks
2. " options for tasks
3. Combine results
4. trim the hamburger
5. take 1st bite
6. take another bite & continue.

Project Scheduling. 30/8

plan driven methods	Agile methods
1. Mapping effort to durat 2. eg of Effort & time distn 3. Work breakdown Stru (WBS) 4. Gantt charts 5. AoN dig & Critical path	1. eg of effort distn 2. eg planning tool - Pivotal Tracker (US, bugs, chores, milestones).

Scheduling in plan driven methods

1. defining proj tasks & milestones
2. Mapping estimated effort to duratn
3. Identifying dependencies
4. Developing a WBS
5. Gant charts
6. Aon dig & Critical path.

2. Mapping eff effort to d

- Effort in functn poi KLOC
- Effort to duratn.
- exp based & heuristi based methods
- Parametric meth lik COCOMO 81 & COC

eg:
assume for MUX -
- functn points ~ 500
- avg productivity per month per de
- # no. of dev

ideal proj duratn
1 dev - 50 person
2 dev - 25 "
50 dev - 1 persn

COCOMO - formu duratn.

$$\boxed{M = a \times E^b}.$$

E = Estimated eff

Project type	a
Organic	2.5
Semi-detached	2.5
Embeded	2.5

- assume mux-core is Semi-detached proj type
with 500 fp, then proj duratn shd be ≈22 mnths.

$= 2.5 \times 500^{0.35} \approx 22$ (app)

1. Organic

- Small team size
- Problm is well understood & solved in past
- tm has nominal exp reg prblm.

2. Semi detached.

- team size, exp, knowlldge lie b/w Organic & Embedded.
- less familiar & diffi to develop compared to Organic one.

eg: Compilers / diff Embedded Sys ~~com~~

3. Embedded.

- ↑ complexity, creativity, exp.
- large team size.
- ↑ exp dev.

eg: if proj is estimated to be 400 kloc (Killo lines of code). ~~Effort~~ time would be?

1. Organic

~~E = 2.4 x(400)~~

Time = 2.5 x

Effort req is 500 fp

1. Organic

Duratn $= 2.5 \times 500^{0.38}$

2. Semi-Detached

Duratn $= 2.5 \times 500^{0.35}$

3. Embeded

Duratn $= 2.5 \times 500^{0.32}$

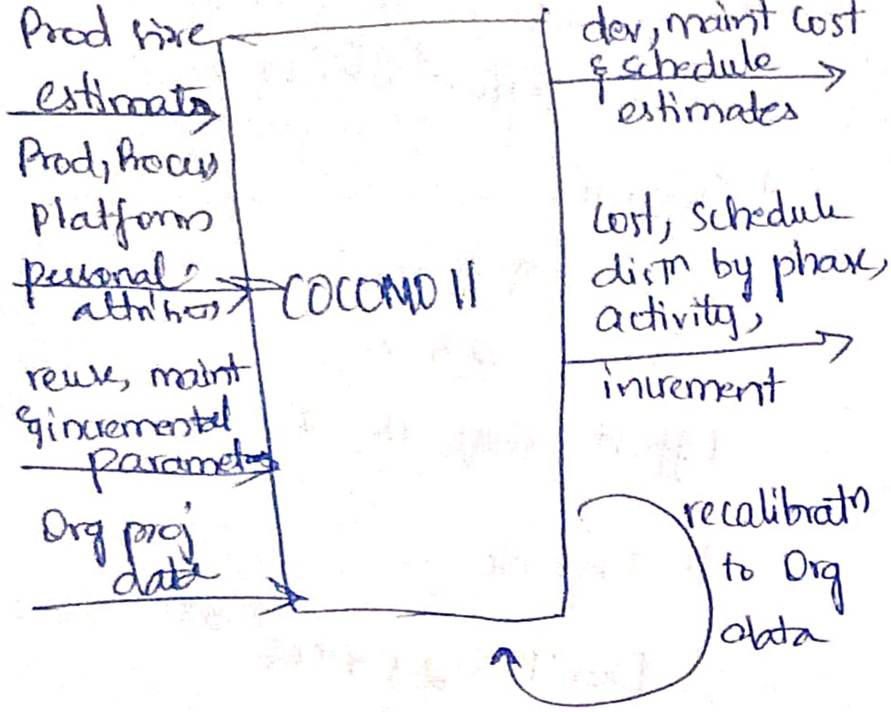
eg of Effort Distn.

requirement → Design → Dev → Verification → Maintenance

Quality & review ↑, Q&r ↑, Q&r ↑, Q&r ↑

requirement	Design	Dev	Verification	Maintenance
15% of total proj effort on capturing, planning	15% effort on high & low level design	35% on coding & unit testing	25% on Sys Integrtn & UAT	10% on release activity & maintenance

COCOMO black box model.



COCOMO II Calculator (img)

Will give results of
Effort, Schedule, Cost
total Equivalent size
EAF (Effort adjustment factor)

Unified Process model.

COCOMO II Cal uses:
acquisitn phase distn as
Inception
Elaboratn
Constructn
Transition.

3. Work Breakdown structure :-

- breaking work in parts & assign the parts to proj team
- anticipate problems might rise & p... soln for them.

+ Project plan
- Created at sta... proj.
- used to comm... how work will b... done to proj tea... Cust & help to assess progress o...

eg: of WBS frag.
3:01: design a phy...

3.01.01
3.01.02
:
3.01.10
(refer ppts).

eg of WBS fragment

Level 3: Sub tasks in the phase.

1. Req Gathering & Analysis

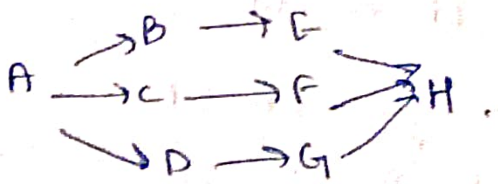
1.1 req Elicitatn
- stk inter
- workshop
- Survey.

1.2 Req Docn
- Functional req Spec (FRS)
- NFRs

1.3 Req review & Validtn.
- Sessions
- Sign-off

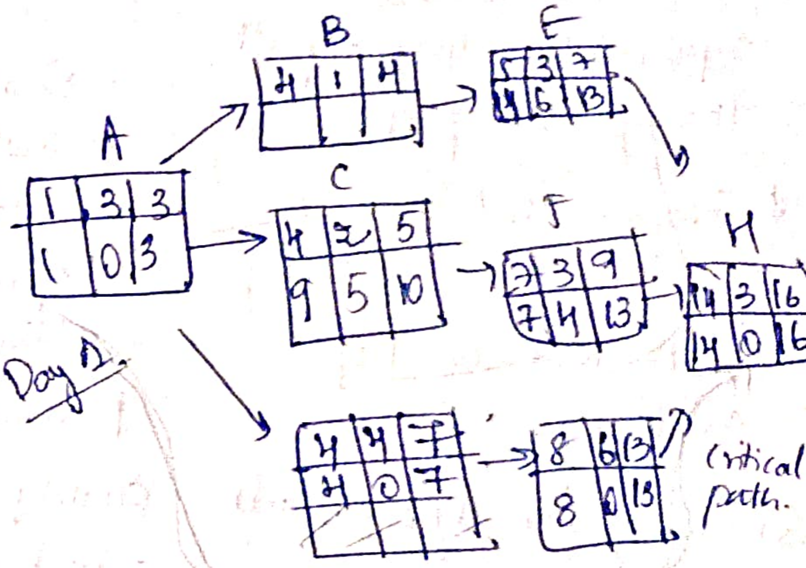
1.4 feasibility study
- tech feasibility
- Budget & resrc feasibility

Activity on Node dig



for every box

Early start	Duratn	Early finish
	Task Name	
Late start	slack	late finish.



$LS = LF - \text{Duratn} + 1$.

$\text{float/slack} = LS - ES$ (or) $LF - EF$

$EF = ES + \text{Duratn}$.

4. Gantt chart

Activity	duratn	depends on	resrc
A	3		SA
B	1	A	SD1
C	2	A	SD2
D	6	B	CD1
E	3	ABC	SA

person doing this

helps finding critical path & duration.

Activity	Duratn (days)	depends on
A	3	
B	1	A
C	2	A
D	4	A
E	3	B
F	3	C
G	6	D
H	3	E,F,G

Answer for this dig is 18 days of duration.

If starting c̄ Node 0. (0th day)

(LS = LF − Duratn)

Fwd Pass gives ES EF

Bwd Pass " LS LF

Float/Slack ~~EF−ES~~ LS−ES
EF−LF

Critical path : Sequence of activities c̄ zero float. representating longest path through the proj

* Compare suitability of Gant charts & AON dig, for ACL MUX Core Proj scheduling.

Scheduling for Agile Methods

- Allocate time for Sprint 0 activities
- Distribution of Effort & time (not uniform across Sprints
- eg of agile management tool.

Scheduling in waterfall

- Upfront planning
- Fixed Timeline
- Dependency driven
- Progress monitoring
- challenges

Scheduling in Scrum

- Iterative planning
- Flexible Timeline
- Sprint Based
- Progress monitoring
- challenges.

Quality Planning.

- Quality manageme
- Quality planning
- identificatn relevant
- Defining associated
- Setting up stde for processes.

Quality.

- Prod chd meet its s

Quality management

1. Quality Planning
2. " Assuranc
3. " Control

1. Quality Planning -

- Setting quality stds developing a plan to a

Prod Vs Process stds:

- design review form - design review conduct
- req doc struct - submission of new code for sys building
- Programming style - Proj plan approval Process
- Proj plan format - change cntrl process
- chang req form - test recording Process

9 9 9 9 9 %
35d 4d 8hr 50 min 5min

= 99.999 %
= 99.99 %
= 99.9 %
= 99 %

} - Availability

Sw Quality Attributes

Safety	modularity
Sec	Complexity
Reliabi	Portability
UsRealibility	Usability

Quality Planning.

- Identifies most significant QA
- defines assesment process in detail for each QA
- indicates std shd be applied & defines new stds as necessary.

QA eg:

Reliability - Probabity of sys working satisfactory in a specific period of time.

Possible measures:

- availability
- Mean time b/w failures: $\frac{\text{total Service time}}{\text{no. of failures}}$
- failure on demand
- Support activity

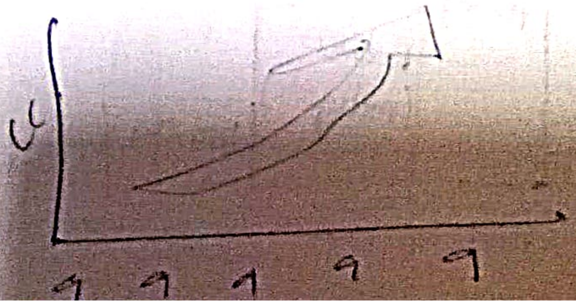
ISO/IEC 25010 std.

Extermall
Internal
Quality

functional suitability

(refer ppt)

Trade offs: Cost & Complexity Vs Avai



ISO/IEC 25010
Prod Quality Model

- Functional suitability
- Perf Efficiency
- Compatibility
- Usability
- Reliability
- Security
- Maintainability
- Portability

Common Prac for Selecting attributes & Metrics:

1. FS (1-2 metrics)
2. Perf eff (2-3 metrics)
3. Compat (1-2 ")
4. Usu (2-3 ")
5. Reliab (2-3 ")
6. Sec (2-3 m)
7. main (1-2)
8. Por (1-2)

typical Quali planning in Waterfall.

1. define QA & metrics
2. develop Quality management plan
3. conduct requirements Validation
4. ~~planning~~ planning testing phases
5. Risk based Quality Prioritization.

Typical Quality planning in Scrum.

1. define Sprint Quality goals
2. Incorpate quality in Backlog refinement
3. implement Continuous testing
4. Conduct sprint reviews for Quality fb
5. Use retrospective for Quality improvement.

* Risk Management :

- Proj can be inherently complex & Uncertai due change in req.
- Effective risk manag. help to adequate issues & en Proj success by ↓ impact -ve risk & ↑ " " +v

1. Risk Identificatn: what c to proj?
2. Risk analysis: which ones really serious?
3. Risk planning: what sh we do?
4. Risk monitoring: has t planning worked?

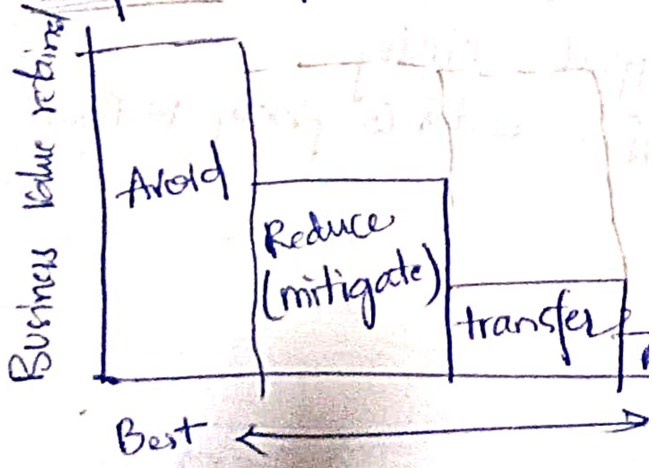
- Risk Identification
  - Use of checklist
  - Brainstorming
  - casual mapping.

(Activity).

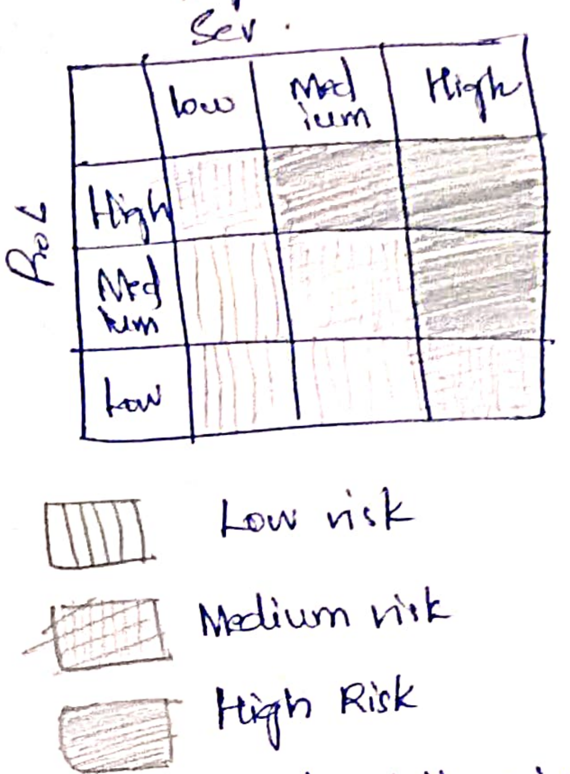
- Risk Mitigation - eg. (refer ppt)

risk | Probability (H/M/L) | impact (H/M/L) | mitig strateg

- Response options



Risk Probability & Severity.



Commonly used risk items in sw dev proj

1. Small proj (<6m, <5tm)
   10-12 risks
2. Medium (6-12m, 5-15tm)
   15-20 risks
3. Large (>12m, >15tm)
   20+ risks

Typical Risk managment in
waterfall   scrum.

risk management tools

1. Ms project
2. Primavera Risk analysis
3. Gloxis
4. Asana & Trello
5. ProofHub.